Inversion and projective methods

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L5-GE1 Level 5 Advanced · Geometry

Inversion, cross-ratio, harmonic division, poles and polars, Brocard's theorem and projective maps, with complete proofs.
By the end you can choose a centre and radius of inversion and translate a problem through the inversion; recognise and prove harmonic ranges, and use poles, polars and Brocard's theorem; use a projective map to send a line to infinity and write the argument rigorously.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Prove that any four points A, B, C, D in the plane satisfy $AB \cdot CD + AD \cdot BC \geq AC \cdot BD$. (Ptolemy's inequality.) Invert about A with radius 1 (defined below). Theorem 1.1 gives $B'C' = BC/(AB \cdot AC)$, and similarly for the pairs C, D and B, D, so the triangle inequality $B'C' + C'D' \geq B'D'$ reads

$$\frac{BC}{AB \cdot AC} + \frac{CD}{AC \cdot AD} \geq \frac{BD}{AB \cdot AD}.$$

Multiplying by $AB \cdot AC \cdot AD$ gives the result. Equality holds if and only if C' lies on segment $B'D'$, and by Theorem 1.2 this happens if and only if A, B, C, D lie on a circle (or a line) with C on the arc BD not containing A.

Power of a point, radical axes, Ceva, Menelaus and directed angles (L4-GE1, L4-GE2) are assumed.

1.1 Inversion

Let ω be a circle with centre O and radius r. The *inverse* of a point $P \neq O$ is the point P' on ray OP with $OP \cdot OP' = r^2$. Inversion is its own inverse, fixes ω pointwise and swaps its inside and outside. We add a point at infinity and say that inversion swaps O and ∞.

Theorem 1.1 (distance formula).
If $P, Q \neq O$ and O, P, Q are not collinear, then triangles OPQ and $OQ'P'$ are similar, and

$$P'Q' = \frac{r^2 \cdot PQ}{OP \cdot OQ}.$$

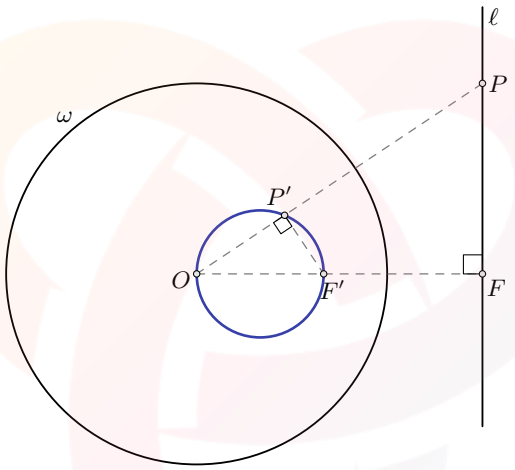
Proof. From $OP \cdot OP' = OQ \cdot OQ'$ we get $OP/OQ = OQ'/OP'$, and the angle at O is common, so $OPQ \sim OQ'P'$ with ratio $OQ'/OP = r^2/(OP \cdot OQ)$. The collinear case is a direct computation. ■

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Theorem 1.2.
(1) A line through O is mapped to itself. (2) A line not through O is mapped to a circle through O, and conversely. (3) A circle not through O is mapped to a circle not through O.

Proof. Statement (1) is in the definition. For (2), let F be the foot of the perpendicular from O to the line ℓ. For P on ℓ, Theorem 1.1 gives $\angle OP'F' = \angle OFP = 90^\circ$, so P' lies on the circle with diameter OF'. Running the argument backwards shows every point of that circle other than O is the image of a point of ℓ. The tangent to this circle at O is parallel to ℓ. For (3), let γ be the circle and $k \neq 0$ the power of O with respect to γ. For P on γ, line OP meets γ again at Q with $\overrightarrow{OP} \cdot \overrightarrow{OQ} = k$ (with $Q = P$ for a tangent). Then $\overrightarrow{OP'} = \frac{r^2}{OP^2}\overrightarrow{OP} = \frac{r^2}{k}\overrightarrow{OQ}$, so P' is the image of Q under the homothety with centre O and ratio r^2/k. Hence γ' is the image of γ under that homothety. ■

The proof of (3) shows that the centre of γ is *not* mapped to the centre of γ'. The two centres are only collinear with O.



Theorem 1.3.
Inversion preserves the angle between two curves (lines or circles) at a common point $P \neq O$. In particular tangent curves are mapped to tangent curves, and orthogonal circles to orthogonal circles.

Proof. Two different circles or lines touch at P if and only if P is their only common point, and inversion is a bijection, so curves touching at $P \neq O$ go to curves touching at P'. Now let γ pass through P with tangent t, and let δ be the circle through O tangent to t at P (the line OP if t passes through O). Then δ' is a line touching γ' at P', parallel to the tangent to δ at O. The tangents to δ at O and at P are mirror images in the perpendicular bisector of OP. So the tangent to γ' at P' makes with line OP the angle opposite to the one t makes, and the angle between two curves through P keeps its size and reverses orientation. ■

Theorem 1.4 (fixed circles).
A circle $\gamma \neq \omega$ is mapped to itself if and only if γ is orthogonal to ω. Every circle through a pair of inverse points $P \neq P'$ is orthogonal to ω.

Proof. By the proof of 1.2(3), γ' is the image of γ under a homothety at O of ratio r^2/k. For $\gamma \neq \omega$ the image equals γ only when the ratio is 1, that is, $k = r^2$ (ratio -1 would need centre O and

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radius r). Now $k = r^2$ says that the tangents from O to γ have length r, that is, their points of contact lie on ω, where the radii of the two circles are then perpendicular. This is orthogonality. A circle through P and P' has $k = OP \cdot OP' = r^2$. ■

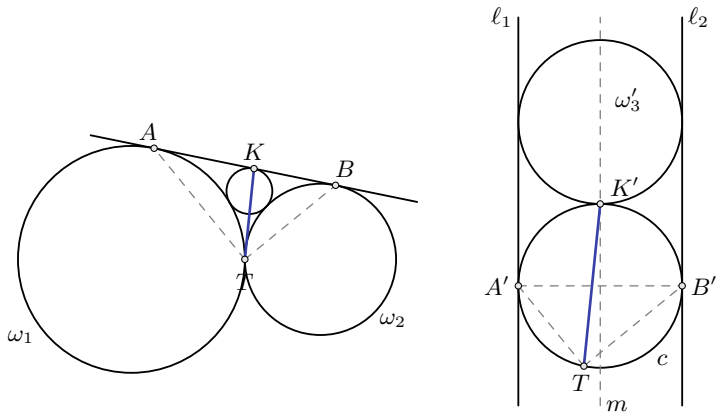
1.2 Choosing the centre and the radius

1. Invert at a point through which several circles pass. They become lines. Two circles touching at the centre of inversion have no other common point, so they become parallel lines.
2. If a product such as $AX \cdot AY$ is fixed, invert at A with radius $\sqrt{AX \cdot AY}$. Then X and Y swap, and every circle through X and Y is mapped to itself.
3. ($\sqrt{bc}$ inversion.) In triangle ABC, invert at A with radius $\sqrt{AB \cdot AC}$ and then reflect in the bisector of angle A. This map swaps B and C, swaps line BC with the circumcircle Γ, and sends each line through A to the isogonal line. For example, the foot D of the altitude goes to the point of Γ on line AO, the antipode A^* of A, and $AD \cdot AA^* = AB \cdot AC$ is the formula $2R \cdot AD = bc$.
4. Before solving, write down what every object becomes.

Example 1.
Circles ω_1 and ω_2 touch externally at T. A common tangent, not through T, touches ω_1 at A and ω_2 at B. A third circle ω_3 touches ω_1 and ω_2, and touches the segment AB at a point K between A and B. Prove that $\angle ATK = \angle KTB = 45°$.

Solution.
Invert at T. The circles ω_1 and ω_2 become parallel lines ℓ_1 and ℓ_2. Line AB does not pass through T, so line AB becomes a circle c through T, touching ℓ_1 at A' and ℓ_2 at B' (Theorem 1.3). Hence $A'B'$ is a diameter of c perpendicular to both lines, and the centre of c lies on the line m midway between ℓ_1 and ℓ_2.
The circle ω_3 does not pass through T. Otherwise the image of ω_3 would be a line parallel to ℓ_1 touching c, which is ℓ_1 or ℓ_2. So ω_3' is a circle touching ℓ_1, ℓ_2 and c. Such a circle has centre on m and the same radius as c, and two different circles of equal radius touch at the midpoint of their centres. Hence the point of contact K' lies on m, the perpendicular bisector of $A'B'$, and K' is the midpoint of an arc $A'B'$ of c.
Inversion maps line AB together with ∞ onto c, and ∞ goes to T. The segment AB avoids ∞, so the image of the segment is the arc $A'B'$ that avoids T. Hence K' is the midpoint of the arc $A'B'$ not containing T, and the inscribed angles $A'TK'$ and $K'TB'$ stand on quarter circles. The points A', B', K' lie on the rays TA, TB, TK, so $\angle ATK = \angle KTB = 45°$.
Remark. Touching circles are awkward and parallel lines are not, so the point of contact is the first centre to try.

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1.3 Cross-ratio and harmonic division

For collinear points A, B, C, D define

$$(A, B; C, D) = \frac{CA}{CB} \div \frac{DA}{DB}$$

with signed lengths. If four lines through a point P meet a line at A, B, C, D, the area formula gives $\frac{CA}{CB} = \frac{PA \sin \angle CPA}{PB \sin \angle CPB}$, and the same for D, so PA and PB cancel and $(A, B; C, D)$ depends only on the four lines (using directed angles for signs). We write $P(A, B; C, D)$ for this common value. **Cross-ratio is preserved by projecting from a point onto another line.** For four points on a circle, $P(A, B; C, D)$ is the same for every P on the circle, by inscribed angles (the line from P to itself is the tangent at P). We call this value $(A, B; C, D)$ on the circle. Since a chord XY equals $2R \sin \angle XPY$, the absolute value is $\frac{CA}{CB} \div \frac{DA}{DB}$ in chord lengths, and the sign is negative when C, D lie on different arcs AB.

We call $A, B; C, D$ *harmonic* if $(A, B; C, D) = -1$, that is, C and D divide AB internally and externally in the same ratio. Then $(C, D; A, B) = -1$ as well. Note that $(A, B; C, \infty) = CA/CB$, so $(A, B; C, \infty) = -1$ if and only if C is the midpoint of AB. Hence a harmonic pencil cuts a line parallel to one of its four lines in a segment and its midpoint.

Theorem 3.1.
(1) If C, D on line AB are the feet of the internal and external bisectors from P in triangle PAB, then $(A, B; C, D) = -1$. Conversely, if $(A, B; C, D) = -1$ and $\angle CPD = 90^\circ$, then PC and PD bisect the angles between PA and PB.
(2) Let cevians AD, BE, CF of triangle ABC be concurrent, and let line EF meet line BC at T. Then $(B, C; D, T) = -1$.
(3) If $(A, B; C, D) = -1$ and M is the midpoint of AB, then $MC \cdot MD = MA^2$.
(4) If points X, Y, Z on a line have coordinates x, y, z measured from a point P of the line, then $(X, Y; P, Z) = -1$ if and only if $\frac{2}{z} = \frac{1}{x} + \frac{1}{y}$.

Statement (1) is the angle bisector theorem, as $\frac{CA}{CB} = \frac{PA}{PB} = \frac{DA}{DB}$ in absolute value, with opposite signs. For the converse, the circle with diameter CD passes through P and is the Apollonius circle of points X with $XA/XB = CA/CB$, so $PA/PB = CA/CB$ and PC is the internal bisector. For (2), Ceva gives $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ and Menelaus for the line EFT gives $\frac{BT}{TC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$ (signed). Dividing, $\frac{BD}{DC} = -\frac{BT}{TC}$. Statements (3) and (4) are short computations. For (3) put $M = 0$, $A = -1$, $B = 1$, and $(A, B; C, D) = -1$ becomes $cd = 1$. For (4), $(X, Y; P, Z) = \frac{x}{y} \cdot \frac{z-y}{z-x} = -1$ rearranges to $z(x + y) = 2xy$.

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Theorem 3.2 (complete quadrilateral).
Let $ABCD$ be a quadrilateral with $Q = AB \cap CD$, $R = AD \cap BC$ and $P = AC \cap BD$. Let line QR meet AC at W and BD at Z, and let line QP meet AD at X. Then

$$(A, C; P, W) = (B, D; P, Z) = (Q, R; W, Z) = (A, D; X, R) = -1.$$

Proof. In triangle RAC the cevians AB, CD, RW concur at Q, and the line BD through two of their feet meets AC at P. So $(A, C; W, P) = -1$ by 3.1(2). The same argument in triangle RBD gives $(B, D; Z, P) = -1$, and in triangle QAD (cevians AC, DB, QX through P) gives $(A, D; X, R) = -1$. Projecting A, C, P, W from B onto line QR gives $(Q, R; Z, W) = -1$. ■

Lemma 3.3 (inversion and cross-ratio).
Consider an inversion with centre O. (1) For points on a line through O, cross-ratios are preserved, where O and ∞ correspond. So if $(A', C'; O, W) = -1$, then the inverse of W is the midpoint of AC. (2) For points on a circle γ through O, cross-ratios are preserved, where O corresponds to the point at infinity of the line γ'.

Proof. (1) On the line, inversion is $x \mapsto r^2/x$, and $\frac{1}{c} - \frac{1}{a} = \frac{a-c}{ac}$ shows that the four factors ac, bc, ad, bd cancel in the cross-ratio. (2) The point X' lies on line OX, so inversion restricted to γ is the projection of γ from O onto γ', and the tangent at O is parallel to γ'. ■

1.4 Poles and polars

The *polar* of a point $P \neq O$ with respect to ω is the line through the inverse P' perpendicular to OP, and P is the *pole* of that line. If P is outside ω and PT is a tangent, then $OT^2 = OP \cdot OP'$ shows that P' is the foot of T on OP, so the polar of P is the chord joining the two points of contact. The polar of a point of ω is the tangent there. With O as origin, the polar of P is the line of points X with $X \cdot P = r^2$ (dot product).

Theorem 4.1 (La Hire).
Q lies on the polar of P if and only if P lies on the polar of Q.
Theorem 4.2. If a line through P (not on ω) meets ω at X, Y and the polar of P at Z, then $(X, Y; P, Z) = -1$.
Theorem 4.3 (harmonic quadrilateral). Let A, B, C, D lie on ω in this order. Then $(A, C; B, D) = -1$ if and only if $AB \cdot CD = BC \cdot DA$. If AC is not a diameter, this happens if and only if BD passes through the pole of AC.

Theorem 4.1 is trivial in coordinates. Both statements say $P \cdot Q = r^2$. For 4.2, take O as origin, $r = 1$, and write the line as $P + tu$ with $|u| = 1$. The parameters t_1, t_2 of X, Y are the roots of $t^2 + 2(P \cdot u)t + |P|^2 - 1 = 0$, and Z has parameter $t_3 = (1 - |P|^2)/(P \cdot u)$. Then $\frac{1}{t_1} + \frac{1}{t_2} = \frac{-2P \cdot u}{|P|^2 - 1} = \frac{2}{t_3}$, which is Theorem 3.1(4). If $P \cdot u = 0$, then $Z = \infty$ and $t_2 = -t_1$, so again $(X, Y; P, Z) = -1$.

For 4.3, the first statement is the chord formula of Section 3. Now let T be the pole of AC and let line TB meet ω again at D_1 and AC at Z. Then $(B, D_1; T, Z) = -1$ by 4.2, and projecting from A onto ω (line AT is the tangent at A) gives $(B, D_1; A, C) = -1$. A harmonic conjugate is unique, so $(A, C; B, D) = -1$ if and only if $D = D_1$. ■

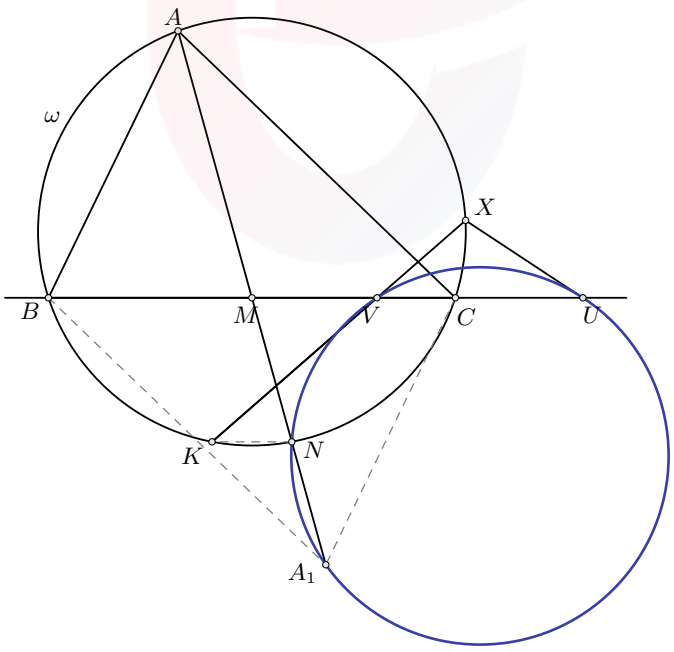
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Example 2.
Let ABC be a triangle with $AB \neq AC$ and $\angle A \neq 90°$, with circumcircle ω, and let M be the midpoint of BC. Line AM meets ω again at N, and the line through N parallel to BC meets ω again at K. For a variable point X of ω, other than A, B, C, K, lines XA and XK meet line BC at U and V. Prove that the circle NUV passes through a fixed point other than N.

Solution.
Step 1. $ABKC$ is a harmonic quadrilateral. Triangles MAB and MCN are similar, as $\angle MAB = \angle MCN$ in ω, and so are MAC and MBN. Hence

$$\frac{AB}{CN} = \frac{MB}{MN} = \frac{MC}{MN} = \frac{AC}{BN}.$$

The reflection in the perpendicular bisector of BC swaps N and K, so $CN = BK$ and $BN = CK$. Thus $AB \cdot KC = BK \cdot CA$. The points N and K lie on the arc BC not containing A, so A, B, K, C are in this order and $(A, K; B, C) = -1$ by Theorem 4.3.
Step 2. Project from X onto line BC. The lines XA, XK, XB, XC meet BC at U, V, B, C, so $(U, V; B, C) = -1$. By Theorem 3.1(3), $MU \cdot MV = MB^2$, with U and V on the same side of M.
Step 3. The power of M gives $MA \cdot MN = MB \cdot MC = MB^2$. Let A_1 be the reflection of A in M, which lies on ray MN. Then $MU \cdot MV = MN \cdot MA_1$, and $A_1 \neq N$ because $MA \neq MB$ when $\angle A \neq 90°$. By the converse of power of a point, U, V, N, A_1 are concyclic. So every circle NUV passes through A_1, the fourth vertex of the parallelogram ABA_1C.
Remark. We want a relation between U and V that does not mention X. Projecting four fixed points of ω from X gives one, and the relation is simplest when the four points are harmonic. That is why we tested A, B, K, C.



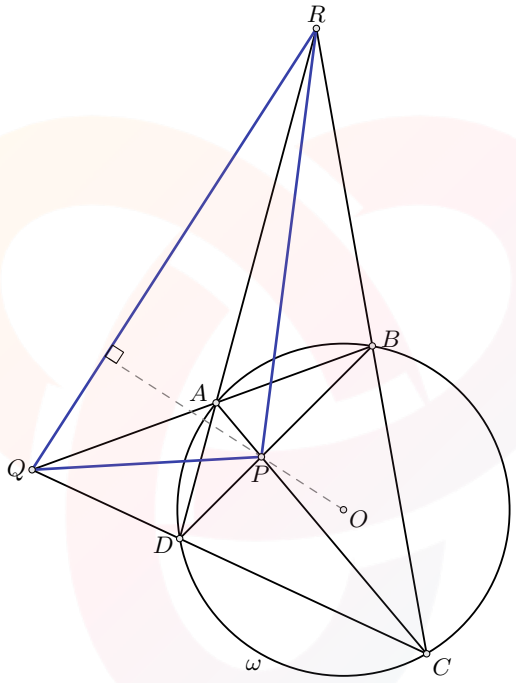
Theorem 4.4 (Brocard).
Let $ABCD$ be inscribed in ω, and suppose $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$ exist. Then QR, RP, PQ are the polars of P, Q, R. Hence O is the orthocentre of triangle PQR.

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Lemma 4.5 (conjugate points). If R lies on the polar of Q and M is the midpoint of QR, then $OM^2 = r^2 + MQ^2$. So the circle with diameter QR is orthogonal to ω, and QR^2 is the sum of the powers of Q and R.

Proof of 4.4. Let QP meet AD at X and BC at Y. By Theorem 3.2, $(A, D; X, R) = -1$, and in the same way $(B, C; Y, R) = -1$. By Theorem 4.2 and the uniqueness of the harmonic conjugate, X and Y lie on the polar of R, so that polar is line $XY = PQ$. Similarly the polar of Q is PR. Now P lies on the polars of Q and R, so by La Hire, Q and R lie on the polar of P, which is therefore QR. A polar is perpendicular to the line joining its pole to O, so $OP \perp QR$, $OQ \perp RP$, $OR \perp PQ$. ■

Proof of 4.5. With O as origin, $Q \cdot R = r^2$. Hence $|M|^2 - MQ^2 = \frac{1}{4}|Q+R|^2 - \frac{1}{4}|Q-R|^2 = Q \cdot R = r^2$, and $|Q-R|^2 = (|Q|^2 - r^2) + (|R|^2 - r^2)$. ■



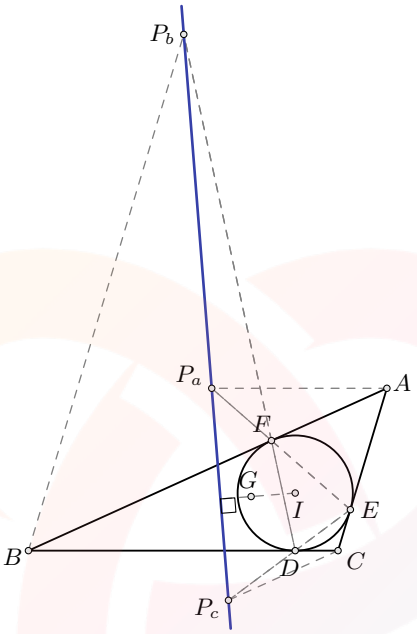
For the incircle of triangle ABC, touching BC, CA, AB at D, E, F, the polars of A, B, C are EF, FD, DE. If EF meets BC at T, then T lies on the polars of A and D, so the polar of T is AD and $IT \perp AD$. Also $(B, C; D, T) = -1$ by 3.1(2), since AD, BE, CF concur by Ceva.

Example 3.
The incircle of a scalene triangle ABC, with centre I, touches BC, CA, AB at D, E, F, and G is the centroid. The line through A parallel to BC meets line EF at P_a, and the points P_b on FD and P_c on DE are defined in the same way. Prove that P_a, P_b, P_c lie on a line perpendicular to IG.

Solution.
Let M be the midpoint of BC and let the median AM meet EF at K. Since M is the midpoint of BC, the lines AB, AC, AM and the parallel to BC through A form a harmonic pencil. Line EF cuts these four lines at F, E, K, P_a, so $(F, E; K, P_a) = -1$.
The point P_a is not on the incircle, and line EF through P_a meets the incircle at E and F. By Theorem 4.2 and the uniqueness of the harmonic conjugate, K lies on the polar of P_a. Also P_a lies on EF, the polar of A, so by La Hire A lies on the polar of P_a. Hence the polar of P_a is

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the median AM.
In the same way the polars of P_b and P_c are the medians from B and C. The three medians pass through G, so by La Hire P_a, P_b, P_c lie on the polar of G. As the triangle is scalene, $G \neq I$, and this polar is a line perpendicular to IG.
Remark. A midpoint and a parallel line make a harmonic pencil. Cutting the pencil by EF, the polar of A, identifies the polar of P_a, and this is the step to remember. Collinear points are concurrent polars, so we look for the polars first.



1.5 Projective maps

Write the point (x, y) as $[x : y : 1]$, where $[x : y : z] = [\lambda x : \lambda y : \lambda z]$ for $\lambda \neq 0$. The points $[x : y : 0]$ form the line at infinity. A *projective transformation* is a map $[v] \mapsto [Mv]$ with M an invertible 3×3 matrix. This map is a bijection, sends lines to lines and preserves cross-ratios (Chapter 9 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads* develops this).

Theorem 5.1.
(1) For any line ℓ there is a projective transformation sending ℓ to the line at infinity. (2) For any circle ω and point P inside ω, there is a projective transformation sending ω to a circle and P to the centre of that circle.

Proof. (1) If ℓ is $ax + by + cz = 0$, take M with third row (a, b, c) and two further rows making M invertible. Then ℓ goes to $z = 0$. (2) Similarities are projective, so we may take ω to be $x^2 + y^2 = z^2$ and $P = [p : 0 : 1]$ with $p \in [0, 1)$. The map $[x : y : z] \mapsto [x - pz : \sqrt{1 - p^2}\, y : z - px]$ multiplies $x^2 + y^2 - z^2$ by $1 - p^2$, so the map sends ω to itself and sends P to $[0 : 0 : 1 - p^2]$, the centre. ■

For a second proof of $(A, D; X, R) = -1$ in Theorem 3.2, send line QR to infinity. Then $ABCD$ becomes a parallelogram with centre P, line QP is parallel to AB, and X is the midpoint of AD while R is at infinity. Only statements about incidence, tangency and cross-ratio survive such a map.

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Common traps.

- Assuming that the centre of a circle is mapped to the centre of its image.
- Forgetting the special cases of inversion, namely lines and circles through the centre, and the centre itself.
- Dropping signs in a cross-ratio, or forgetting that a harmonic conjugate can be the point at infinity.
- Using a projective map to deduce a statement about lengths, angles or perpendicularity.

1.6 How to spot this in a contest

1. Several circles through one point, or two circles touching at a point. Invert there.
2. Equal products of lengths from one vertex, such as $AX \cdot AB = AY \cdot AC$. Choose the radius so that these points swap, and ask what else swaps.
3. Tangents from a point together with a secant, or a cyclic quadrilateral with the intersections of opposite sides. Think of polars, harmonic quadrilaterals and Brocard.
4. A midpoint is a harmonic conjugate of a point at infinity. Project the midpoint, or invert, and look for the harmonic range.
5. A statement with only lines, intersections and one circle is projective. Move a point to the centre or a line to infinity.

Chapter 4 of my book *Beyond the Incenter* treats cross-ratios at greater length, including harmonic bundles, perspectivity and Pascal's theorem.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ A circle ω_2 of radius r_2 lies inside a circle ω_1 of radius r_1 and touches ω_1 at T. A variable circle γ through T meets ω_1 again at P and ω_2 again at Q. The tangents to γ at P and Q meet at X, and line TX meets γ again at M. Prove that M lies on a fixed circle of radius $\frac{2r_1r_2}{r_1+r_2}$.

Problem 2 ⋆⋆⋆ The incircle of a scalene triangle ABC, with centre I, touches BC, CA, AB at D, E, F. The line through B parallel to CA meets line FD at P, and the line through C parallel to AB meets line DE at Q. Prove that the lines EP, FQ and DI are concurrent.

Problem 3 ⋆⋆⋆⋆ Let ABC be a triangle with $AB \neq AC$. The incircle ω, with centre I, touches BC at D, and the A-excircle, with centre I_A, touches BC at D_a. The bisector of angle A meets BC at L. A circle through L and D_a touches ω at a point N. Prove that line NI_A passes through the point of ω diametrically opposite D.

Problem 4 ⋆⋆⋆⋆ Let $ABCD$ be a convex quadrilateral with no two sides parallel, whose diagonals meet at P, the midpoint of neither diagonal. Circles PAB and PCD meet again at X, and

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circles PBC and PDA meet again at Y. Suppose that XY is not a diameter of the circle PXY. Prove that the tangents to the circle PXY at X and Y meet on the line through the midpoints of AC and BD.

Problem 5 ⋆⋆⋆⋆ Let ABC be a triangle. Let K and L be the feet of the perpendiculars from B and C to the internal bisector of angle A. Let X be the foot of the perpendicular from L to line AB, and let Y be the foot of the perpendicular from K to line AC. Prove that the circle AXY touches both the incircle and the A-excircle of triangle ABC.

Problem 6 ⋆⋆⋆⋆ Let ABC be a triangle with circumcircle Γ, and let D be a point on side BC, other than B and C, such that AD is not perpendicular to BC. The circle ABD meets line AC again at $E \neq A$, and the circle ACD meets line AB again at $F \neq A$. Line AD meets Γ again at G, and the perpendicular bisector of AG meets line BC at Z. Prove that ZA is tangent to the circle AEF.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 2002/03 Problem 2** ⋆⋆
   Find two similar triangles sharing the angle at A. Then compare AP squared with AD times AB.
2. **BMO2 2017/18 Problem 1** ⋆⋆
   Invert at B with power BA times BP and ask where M goes.
3. **BMO2 2013/14 Problem 4** ⋆⋆⋆
   Find the direction of the tangent at P to circle BCP.
4. **IMO 2003 Problem 4** ⋆⋆⋆
   Turn both conditions into statements about DA/DC and BA/BC.
5. **IMO 2010 Problem 4** ⋆⋆⋆
   The circle centred at S through C is orthogonal to the circumcircle. This circle is an Apollonius circle.
6. **APMO 2013 Problem 5** ⋆⋆⋆
   Find two harmonic quadrilaterals. Then project them from points of the circle onto a common line.
7. **IMO 1996 Problem 2** ⋆⋆⋆
   Invert at A, or use the inversion distance formula, to turn the angle condition into a length condition.
8. **IMO 1985 Problem 5** ⋆⋆⋆
   Brocard for the cyclic quadrilateral AKNC gives a line through B perpendicular to a line through O. Then show that M is the foot.
9. **USAMO 2015 Problem 2** ⋆⋆⋆
   Show that AX times AS is constant. Inversion at A then swaps X and S.
10. **IMO 2015 Problem 3** ⋆⋆⋆⋆
    Invert at H with the negative power that swaps the circumcircle and the nine-point circle.
11. **USAMO 2017 Problem 3** ⋆⋆⋆⋆
    Find an inversion centred at S that swaps B and C and also swaps K and M.

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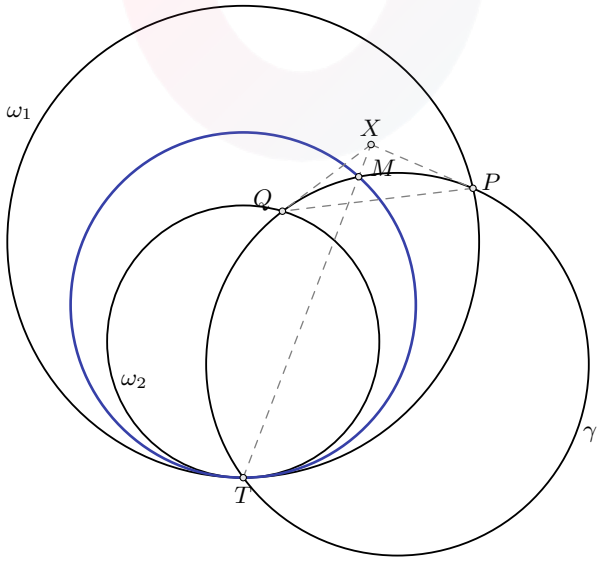
4 Hints

1. What kind of quadrilateral is $TPMQ$? Then invert at T and ask what a harmonic quadrilateral with a vertex at the centre of inversion becomes.
2. Find the polars of P and Q as in Example 3. Then find the poles of EP and FQ, and ask which line joins them.
3. The range $(A, L; I, I_A)$ is harmonic. Project this range onto BC from the point of ω opposite D. Then look at the point where the common tangent at N meets BC.
4. Invert at P. Where do the midpoints of the diagonals go, and what do Theorem 3.2 and Lemma 3.3 say about them?
5. Compute $AX \cdot AB$ and $AY \cdot AC$. Then find an inversion centred at A that moves the incircle to another circle touching lines AB and AC.
6. Apply the $\sqrt{bc}$ inversion. Circle AEF becomes a line through two points of a complete quadrilateral inscribed in Γ, and Brocard gives a line perpendicular to that line. Then look at the circle with centre Z through A.

5 Solutions

Problem 1

A circle ω_2 of radius r_2 lies inside a circle ω_1 of radius r_1 and touches ω_1 at T. A variable circle γ through T meets ω_1 again at P and ω_2 again at Q. The tangents to γ at P and Q meet at X, and line TX meets γ again at M. Prove that M lies on a fixed circle of radius $\frac{2r_1r_2}{r_1+r_2}$.



The chord TM of γ passes through X, the pole of PQ with respect to γ. By Theorem 4.3, $(P, Q; T, M) = -1$ on γ.

Invert at T with radius 1. The circles ω_1 and ω_2 have no common point other than T, so they become parallel lines ℓ_1 and ℓ_2, both perpendicular to the line of centres. The point of ω_i opposite

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T is at distance $2r_i$ from T, so ℓ_i is at distance $\frac{1}{2r_i}$ from T, and the two lines lie on the same side of T. The circle γ becomes a line γ', which meets ℓ_1 at P' and ℓ_2 at Q'.

By Lemma 3.3(2), $(P', Q'; \infty, M') = -1$, so M' is the midpoint of $P'Q'$. Hence M' lies on the line m parallel to ℓ_1 and midway between ℓ_1 and ℓ_2, whose distance from T is

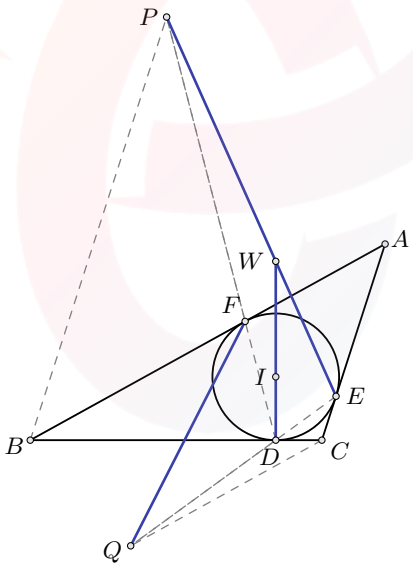
$$d = \frac{1}{2}\left(\frac{1}{2r_1} + \frac{1}{2r_2}\right) = \frac{r_1 + r_2}{4r_1 r_2}.$$

The line m does not depend on γ and does not pass through T. By Theorem 1.2(2) the inverse of m is the circle on diameter TF', where F is the foot of the perpendicular from T to m and $TF' = 1/d$. So M lies on this fixed circle, which has radius $\frac{1}{2d} = \frac{2r_1 r_2}{r_1 + r_2}$ and touches ω_1 and ω_2 at T. ■

Remark. The tangents at P and Q and the line through their meeting point give a harmonic quadrilateral. With one vertex at the centre of inversion, the harmonic condition becomes a midpoint condition, and the two circles become parallel lines.

Problem 2

The incircle of a scalene triangle ABC, with centre I, touches BC, CA, AB at D, E, F. The line through B parallel to CA meets line FD at P, and the line through C parallel to AB meets line DE at Q. Prove that the lines EP, FQ and DI are concurrent.



All poles and polars are taken with respect to the incircle. Let M_b and M_c be the midpoints of CA and AB.

Step 1. The polar of P is the median BM_b. The lines BC, BA, BM_b and the parallel to CA through B form a harmonic pencil, because M_b is the midpoint of CA. Line FD cuts them at D, F, a point K and P, so $(D, F; K, P) = -1$. The point P is not on the incircle, so by Theorem 4.2 the point K lies on the polar of P. Also P lies on FD, the polar of B, so B lies on the polar of P. Hence the polar of P is line BK, the median BM_b. In the same way the polar of Q is the median CM_c.

Step 2. The polar of M_b is EP. The point M_b lies on the polar of P, so P lies on the polar of M_b. The point M_b also lies on CA, the tangent at E, so E lies on the polar of M_b. Since $P \neq E$, the polar of M_b is line EP. In the same way the polar of M_c is line FQ.

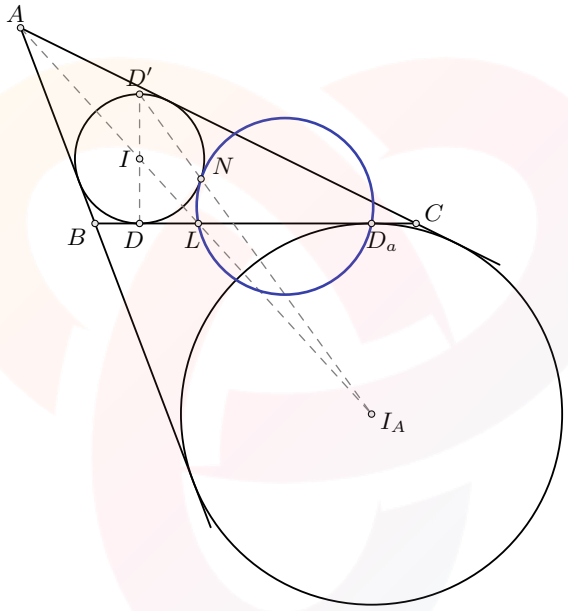
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Step 3. The midline M_bM_c does not pass through I. Its distance from BC is half the altitude h_a, and $h_a = 2r$ would give $a = s$ in $ah_a = 2rs$, which is impossible. So M_bM_c has a pole W, and by La Hire W lies on the polars EP and FQ of M_b and M_c. By the definition of a polar, $IW \perp M_bM_c$. As $M_bM_c \parallel BC$ and $ID \perp BC$, the point W lies on line DI. Thus EP, FQ and DI meet at W. ■

Remark. Three lines are concurrent when their poles are collinear. Two of the poles are midpoints of sides, which is why the midline and the direction of BC enter. The same argument shows that in Example 3 the polar of the midpoint M of BC is DP_a, so $DP_a \perp IM$.

Problem 3

Let ABC be a triangle with $AB \neq AC$. The incircle ω, with centre I, touches BC at D, and the A-excircle, with centre I_A, touches BC at D_a. The bisector of angle A meets BC at L. A circle through L and D_a touches ω at a point N. Prove that line NI_A passes through the point of ω diametrically opposite D.



Let D' be the point of ω diametrically opposite D, and let c be the circle through L, D_a and N. Since $AB \neq AC$, the points D, L, D_a are different and D' is not on line AI.

Step 1. A harmonic range on BC. In triangle ABL the lines BI and BI_A are the internal and external bisectors at B, so $(A, L; I, I_A) = -1$ by Theorem 3.1(1). The homothety with centre A taking ω to the excircle takes the tangent to ω at D', which is parallel to BC, to the tangent BC of the excircle. So A, D', D_a are collinear. The points D' and I_A are on opposite sides of BC, so line $D'I_A$ meets BC at a point Z. Projecting from D' onto BC sends A, L, I, I_A to D_a, L, D, Z. Hence

$$(D_a, L; D, Z) = -1.$$

Step 2. The common tangent. Let t be the common tangent of c and ω at N. First $N \neq D$, as a circle touching BC at D cannot meet BC again at L. Next t is not parallel to BC. Otherwise $N = D'$ and the centre of c is on line DD', so DD' is the perpendicular bisector of the chord LD_a and D is the midpoint of LD_a. Then the harmonic conjugate Z of D would be at infinity, which is false. So t meets BC at a point Z_0.

The tangents from Z_0 to ω give $Z_0N = Z_0D$, and the power of Z_0 with respect to c gives $Z_0N^2 = Z_0L \cdot Z_0D_a$ as a signed product. Hence $Z_0L \cdot Z_0D_a = Z_0D^2$. Let Z_1 be the reflection of D in Z_0.

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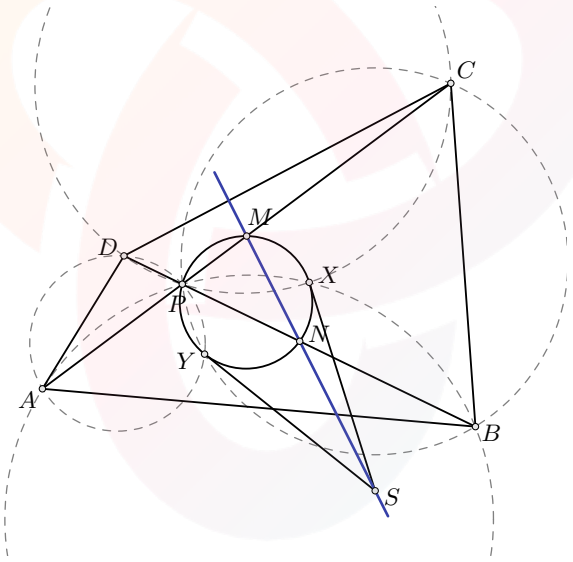
The computation in the proof of Theorem 3.1(3) is reversible, so $(D, Z_1; L, D_a) = -1$. A harmonic conjugate is unique, so $Z_1 = Z$.

Step 3. Now $Z_0D = Z_0N = Z_0Z$, so N lies on the circle with diameter DZ and $\angle DNZ = 90°$. Also $\angle DND' = 90°$, because DD' is a diameter of ω and $N \neq D, D'$. Both NZ and ND' are perpendicular to ND at N, so N, D', Z are collinear. Therefore N lies on line $D'Z$, which is line $D'I_A$. ■

Remark. A circle through two points of a tangent line BC touching ω at N gives $Z_0D^2 = Z_0L \cdot Z_0D_a$, which is equivalent to a harmonic range. So we want a harmonic range on BC containing D, L and D_a. The range $(A, L; I, I_A)$ is the only one in sight, and the centre of projection must send I to D and A to D_a, which forces D'.

Problem 4

Let $ABCD$ be a convex quadrilateral with no two sides parallel, whose diagonals meet at P, the midpoint of neither diagonal. Circles PAB and PCD meet again at X, and circles PBC and PDA meet again at Y. Suppose that XY is not a diameter of the circle PXY. Prove that the tangents to the circle PXY at X and Y meet on the line through the midpoints of AC and BD.



Let M and N be the midpoints of AC and BD. They are different points, since otherwise $ABCD$ would be a parallelogram.

Step 1. The inverted picture. Four circles pass through P, so we invert at P with radius 1. The points A', C' lie on line AC and B', D' on line BD, and the four circles become the lines $A'B'$, $C'D'$, $B'C'$, $D'A'$. If $A'B' \parallel C'D'$, then $PA'/PC' = PB'/PD'$, so $PC/PA = PD/PB$ and $AB \parallel CD$. Hence $X' = A'B' \cap C'D'$ exists, and similarly $Y' = B'C' \cap D'A'$.

Step 2. The midpoints. Apply Theorem 3.2 to the quadrilateral $A'B'C'D'$. Line $X'Y'$ meets $A'C'$ at a point W and $B'D'$ at a point Z with

$$(A', C'; P, W) = (B', D'; P, Z) = (X', Y'; W, Z) = -1.$$

By Lemma 3.3(1) the inverse of W is the midpoint of AC, so $W = M'$, and similarly $Z = N'$. Here $W \neq P$, so line $X'Y'$ does not pass through P, and the inverse of this line is the circle PXY. Hence M and N lie on the circle PXY.

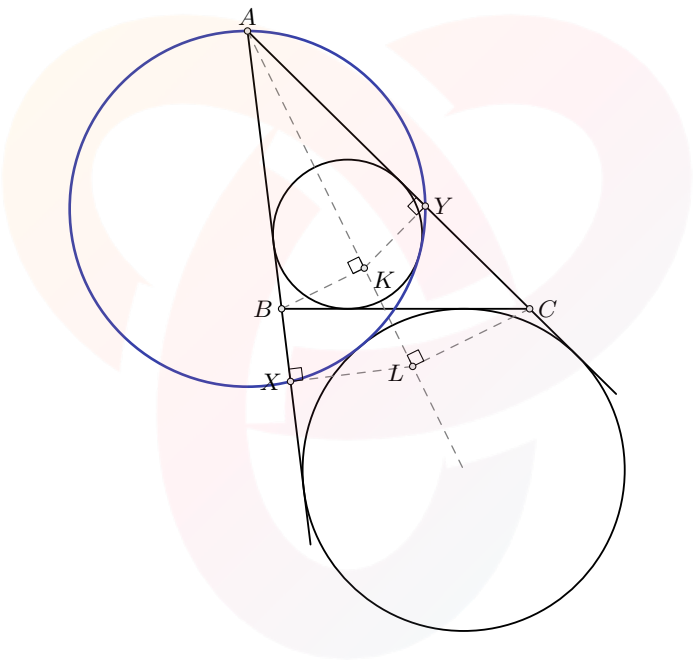
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Step 3. A harmonic quadrilateral. By Lemma 3.3(2), applied to the circle PXY and the line $X'Y'$, we get $(X, Y; M, N) = (X', Y'; M', N') = -1$ on the circle PXY. So M and N lie on different arcs XY and $XMYN$ is a harmonic quadrilateral. As XY is not a diameter, Theorem 4.3 shows that MN passes through the pole of XY, where the tangents at X and Y meet. ■

Remark. The midpoint of AC is the harmonic conjugate of ∞, and the inversion at P swaps ∞ with P. So the midpoints become the harmonic conjugates of P, which a complete quadrilateral supplies. Nothing in the statement says that M and N lie on the circle PXY, and that is the hard step. The line MN is the Newton line of the quadrilateral.

Problem 5

Let ABC be a triangle. Let K and L be the feet of the perpendiculars from B and C to the internal bisector of angle A. Let X be the foot of the perpendicular from L to line AB, and let Y be the foot of the perpendicular from K to line AC. Prove that the circle AXY touches both the incircle and the A-excircle of triangle ABC.



Write a, b, c for the sides, s for the semiperimeter and $\theta = \frac{1}{2}\angle A$. Since θ is acute, K and L lie on the bisector ray from A, with $AK = c\cos\theta$ and $AL = b\cos\theta$. For the same reason X lies on ray AB and Y on ray AC, with

$$AX = AL\cos\theta = b\cos^2\theta, \qquad AY = AK\cos\theta = c\cos^2\theta.$$

Hence $AX \cdot AB = AY \cdot AC = bc\cos^2\theta$. By the cosine rule,

$$bc\cos^2\theta = \tfrac{1}{2}bc(1+\cos A) = \tfrac{1}{4}\left((b+c)^2 - a^2\right) = s(s-a).$$

Let the incircle ω touch AB, AC at F, E and the A-excircle ω_a touch lines AB, AC at F_a, E_a. These four points lie on the rays AB and AC, with $AE = AF = s - a$ and $AE_a = AF_a = s$. Invert at A with radius $\sqrt{s(s-a)}$. This inversion swaps B with X, C with Y, F with F_a and E with E_a.

Lines AB and AC are mapped to themselves. The incircle does not pass through A and touches AB at F and AC at E, so by Theorem 1.3 the image of ω is a circle touching AB at F_a and AC at E_a.

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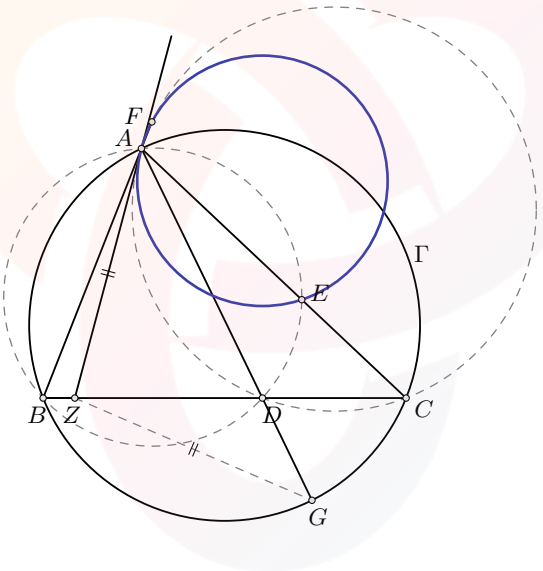
Only one circle does this, because its centre is on the perpendiculars to AB at F_a and to AC at E_a. So the inversion swaps ω and ω_a.

Line BC does not pass through A, so the image of line BC is the circle through A, X and Y. Line BC touches both ω and ω_a. By Theorem 1.3, circle AXY touches the images ω_a and ω. ■

Remark. The products $AX \cdot AB$ and $AY \cdot AC$ are equal, so there is an inversion at A turning circle AXY into line BC, and BC touches both circles. The only question left is whether this inversion respects the pair of circles, which is the identity $AE \cdot AE_a = s(s-a)$. The circle AXY touches ω on the line joining A to the point where ω_a touches BC, and touches ω_a on line AD, where D is the point where ω touches BC.

Problem 6

Let ABC be a triangle with circumcircle Γ, and let D be a point on side BC, other than B and C, such that AD is not perpendicular to BC. The circle ABD meets line AC again at $E \neq A$, and the circle ACD meets line AB again at $F \neq A$. Line AD meets Γ again at G, and the perpendicular bisector of AG meets line BC at Z. Prove that ZA is tangent to the circle AEF.



Let Φ be the $\sqrt{bc}$ inversion of Section 2, that is, the inversion at A with radius $\sqrt{AB \cdot AC}$ followed by the reflection ρ in the bisector of angle A. Write $X^* = \Phi(X)$, and let O be the centre of Γ.

Step 1. The images. Since D lies on side BC, the point D^* lies on Γ, on the arc BC not containing A, and line AD^* is the reflection of line AD. The point G lies on Γ and on line AD, so $G^* = AD^* \cap BC$. The circle ABD becomes the line CD^* and line AC becomes line AB, so $E^* = AB \cap CD^*$. In the same way $F^* = AC \cap BD^*$. These are finite points because $E, F \neq A$, and circle AEF becomes the line E^*F^*.

Step 2. Brocard. The points A, B, D^*, C lie on Γ in this order. In the quadrilateral ABD^*C the diagonals AD^* and BC meet at G^*, and the pairs of opposite sides meet at E^* and F^*. Also $G^* \neq O$, since otherwise both diagonals would be diameters, ABD^*C would be a rectangle, and E^* would not exist. By Brocard's theorem E^*F^* is the polar of G^*, so $OG^* \perp E^*F^*$.

Step 3. The circle with centre Z. The perpendicular bisector of AG is not parallel to BC, so Z exists. Let δ be the circle with centre Z through A and G. Its centre lies on BC, so δ also passes through the reflection A_1 of A in BC. By Section 2, Φ sends the foot of the altitude from A to the antipode

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of A in Γ. The point A_1 is on the same ray from A at twice the distance, so A_1^* is on ray AO at half the distance of the antipode, that is, $A_1^* = O$. As δ passes through A, the image of δ is a line, and this line contains G^* and O. Hence $\Phi(\delta)$ is line OG^*.

Step 4. Tangents at A. By the proof of Theorem 1.2(2), a circle through A is inverted to a line parallel to its tangent at A. So ρ maps the tangent at A to circle AEF to a line parallel to E^*F^*, and maps the tangent at A to δ to a line parallel to OG^*. By Step 2 these two image lines are perpendicular, and a reflection preserves perpendicularity. Therefore the tangents at A to circle AEF and to δ are perpendicular. The tangent to δ at A is perpendicular to the radius ZA, so the tangent to circle AEF at A is the line ZA. ■

Remark. Circles ABD and ACD pass through A and through B or C, so Φ turns them into lines through C and B, and four points on Γ with all their joining lines appear. The condition $ZA = ZG$ names a circle through A orthogonal to BC, and Φ turns orthogonality to BC into orthogonality to Γ, which means a line through O.